

Intelligent Traffic Signal Controller ET-GSC100 Datasheet

- Product Type: Large Series Modular Intelligent Traffic Signal Controller
- Model: ET-GSC100 (Large)
- Doc Type: Datasheet
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Application: Urban arterials, large-scale intersections, multi-phase complex intersections, bus lanes, hospital/fire station adjacent key intersections

1. Product Overview

The ET-GSC100 Large Series is a high-performance, centralized coordinated modular traffic signal controller, engineered to global industrial standards for 24/7 reliable outdoor operation in complex traffic scenarios. Equipped with a high-performance ARM processor and Linux multi-tasking operating system, it supports full-scenario real-time traffic optimization, rich control modes, and seamless integration with radar/vision/coil and other vehicle/pedestrian detection systems. It is the core control device for urban arterial green wave coordination, multi-phase complex intersection management, and key traffic node priority control, delivering stable and efficient performance for global smart city and urban road traffic projects.

2. Key Features

- Modular plug-in chassis structure (main control/lamp control/detection boards independently removable) for easy maintenance and on-demand expansion
- Industrial-grade components with high stability, strong anti-interference, and robust protection for harsh outdoor environments
- Dual control mode: optional 12" touchscreen local HMI + upper platform remote control (support standard protocol connection)
- Comprehensive fault detection & automatic safety protection, hardware yellow flash trigger for fault status
- Wide compatibility with IO/RS422/485/RJ45 interface detectors (inductive loop, magnetic, video, radar, radar-vision integrated)
- Support for self-learning/protocol-based/pulse-type countdown timers and variable lane signs
- Open architecture, full proprietary protocol disclosure, compatible with third-party optimization algorithms
- GNSS (GPS/BeiDou) supported for cable-free green wave coordination of adjacent intersections
- Professional bus/emergency vehicle priority control, dedicated for key traffic nodes with high demand for traffic efficiency

3. Core Technical Specifications

Parameter	Specification
Power Supply	176-264VAC, 25A
Maximum Power	4500W

Consumption	
Signal Lamp Groups	32 groups
Lamp Output Terminals	96 terminals (5A / 220VAC per terminal)
Detector Inputs	32 isolated IO channels; support RS422/485/RJ45 extended detection access
Leakage Detection	32 independent channels
Communication Interfaces	2×Gigabit Ethernet (RJ45), 1×RS232, 1×RS485/RS422, 2×USB Device, 1×DB9 serial
Insulation Resistance	≥10MΩ (normal condition)
Protection Functions	Lightning protection, leakage protection, short-circuit protection, red-green conflict protection
Operating Temperature	-40°C ~ +70°C
IP Rating	IP65 (outdoor enclosure)
Cabinet Structure	19-inch standard rack-mount compact cabinet (front & rear double doors)
Additional Accessory	Manual operation panel + independent backup power door

4. Supported Control Modes

- Multi-period fixed timing control
- Pedestrian request control
- Actuated control (vehicle detection-based green time adjustment)
- Single-point adaptive control
- Single-point optimization control (traffic volume-based cycle/green ratio adjustment)
- Cable-free coordinated control (GNSS-based green wave coordination)
- Bus priority control (green time extension/red time shortening)
- Emergency override control (immediate green phase switch for emergency vehicles)
- Manual panel step control

5. Safety & Fault Protection Functions

- Real-time red-green conflict detection: immediate hardware yellow flash trigger for same lamp group red/green simultaneous on
- Phase directional red light full-off fault detection: automatic hardware yellow flash for complete red light failure
- Green conflict pre-check: automatic phase table verification during configuration to avoid traffic movement conflicts
- Fault status automatic response: one-click hardware yellow flash/all-red mode for emergency fault handling
- Real-time device status monitoring & remote alarm: real-time upload of operating status, abnormal alarm push
- Parameter consistency verification: periodic parameter upload to ensure platform and device settings match
- Comprehensive circuit protection: lightning/leakage/short-circuit protection for all input/output terminals

6. Mechanical & Environmental Characteristics

- Rugged industrial-grade metal enclosure, compact design for space-saving installation
- Front & rear double door access, separate manual operation panel and backup power door for easy on-site operation
- Reinforced anti-corrosion and dustproof design, adapted to outdoor harsh environments (rain, dust, extreme temperature)
- Stable performance under extreme temperature (-40°C ~ +70°C) and high humidity conditions
- Floor-standing mounting design, compliant with 19-inch standard rack-mount specifications for standardized deployment
- Vibration and shock resistance, suitable for long-term operation in urban road side and intersection environments

7. Detector Integration Compatibility

The ET-GSC100 Large Series supports flexible integration with multiple detectors via three interface types, with one-to-one detector-parameter table mapping for plug-and-play configuration:

- Discrete Input (IO): Inductive loop detectors, magnetic detectors, pedestrian push buttons
- RS422/485: Protocol-based radar detectors, video detectors, radar-vision integrated sensors
- RJ45 (Gigabit Ethernet): Network-based intelligent detection devices, traffic data collection terminals

8. Typical Application Scenarios

- Urban arterial roads: green wave coordination via cable-free control, improve overall road traffic efficiency
- Large multi-phase complex intersections: multi-mode adaptive control based on real-time traffic volume
- Dedicated bus lanes: professional bus priority control, ensure public transport travel efficiency
- Key intersections near hospitals/fire stations: emergency vehicle override control, clear rescue passage
- Core urban business district intersections: single-point optimization control, reduce vehicle queuing and stops
- Smart city traffic systems: seamless connection with upper traffic control platforms (1049.2/20999/NTCIP) for centralized management

9. Appearance

(Images shown for reference; actual configuration subject to delivery)



Remark: Specifications and model are for illustrative purposes only. A full range of models is available to meet different project requirements.